

# Polarization mode converters in ZnO-diffused LiNbO<sub>3</sub> waveguides

James E. Toney,\* Jason Retz, Andrea Pollick, Vincent E. Stenger, and Sri Sriram  
SRICO, Inc., Columbus, Ohio, United States

**Abstract.** The linear electro-optic effect in lithium niobate is capable of realizing a variety of polarization transformations, including TE-to-TM and left-to-right circular polarization conversion. Several types of devices have been demonstrated in Ti-diffused waveguides. LiNbO<sub>3</sub> devices have historically been based on either Ti-diffused or proton-exchanged waveguides. The proton-exchanged waveguides only guide light polarized along the optic axis, and therefore, are not applicable to polarization transforming devices. Zinc oxide diffusion is an alternative waveguide fabrication technology that guides both e- and o-waves with much higher power-handling capability than Ti:LiNbO<sub>3</sub> waveguides. ZnO:LiNbO<sub>3</sub> waveguides exhibit a highly circular mode field with lower anisotropy than Ti-diffused waveguides. We report the design, fabrication, and testing of ZnO:LiNbO<sub>3</sub> devices for polarization mode conversion. © 2019 Society of Photo-Optical Instrumentation Engineers (SPIE) [DOI: 10.1117/1.OE.58.8.082420]

Keywords: lithium niobate; polarization control; electro-optic; TE-to-TM conversion.

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## 1 Introduction

Lithium niobate is the most widely used electro-optic material at telecommunication wavelengths, due to its strong electro-optic effect, availability in large-area wafers, and well-developed fabrication technology. Most commercially available devices use optical waveguides based on titanium diffusion or proton exchange. Zinc oxide (ZnO) diffusion is a less widely used waveguide fabrication technology that has received recent attention based on its high power-handling capability and ability to guide orthogonal polarizations.

Polarization transforming devices in Ti:LiNbO<sub>3</sub> waveguides have been reported in the literature<sup>1–6</sup> and are commercially available. This paper reviews the theory of operation of TE-to-TM conversion and other polarization control devices and reports on the design, fabrication, and demonstration of TE–TM mode conversion devices in ZnO:LiNbO<sub>3</sub> waveguides.

## 2 Theory of Operation

This section reviews the most important aspects of the theory of the linear electro-optic effect and its application to polarization transformations. The operation of each device depends on the crystal orientation and direction of the applied field. Figure 1 shows the three configurations discussed in this paper. In Fig. 1(a), the modulating voltage is applied to the center electrode of a coplanar waveguide (CPW) in an *x*-cut crystal. Since the waveguide (illustrated by the dotted line) is beneath the center electrode, the dominant electric field component in the waveguide region is along the crystal's *x*-axis. This enables conversion between TE and TM modes, as discussed in Sec. 2.2. Figure 1(b) shows the same electrode configuration but driven differentially to produce a field along the *y*-axis. A combination of *x*- and *y*-directed RF fields produces a rotating waveplate effect for frequency shifting.<sup>7</sup> Figure 1(c) shows a *z*-cut device with the waveguide positioned between coplanar electrodes, so

that the electric field is along the crystal's *y*-axis. The region between the electrodes is periodically poled, which produces TE–TM mode conversion in a narrow band of wavelengths, as discussed in Sec. 2.3.

### 2.1 Electro-Optic Effect in Lithium Niobate

In most LiNbO<sub>3</sub> devices, the modulating electric field is applied along the crystal's *z*-axis, which takes advantage of the largest electro-optic coefficient *r*<sub>33</sub>; this geometry simplifies the analysis, since a *z*-polarized field does not produce a rotation of the crystal's principal axes. The change in the refractive index for extraordinary (*z*-polarized) waves depends on the applied field as

$$\Delta n_e = -\frac{1}{2}n_e^3 r_{33} E_z. \quad (1)$$

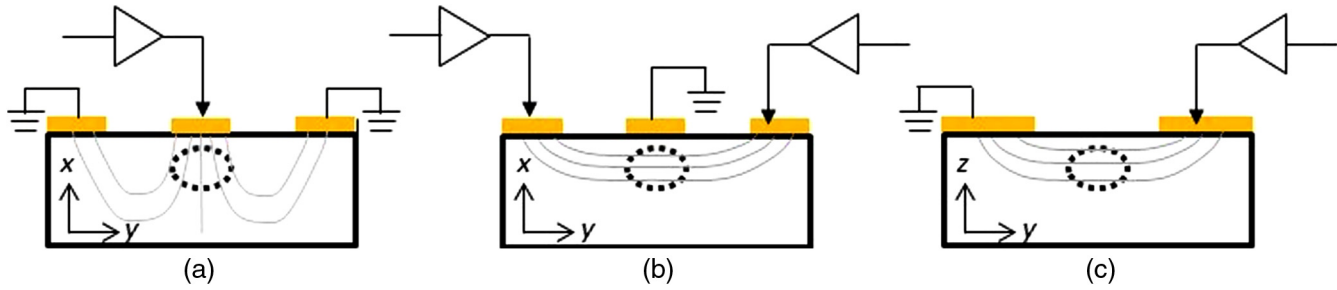
For ordinary waves, *n<sub>e</sub>* is replaced by *n<sub>o</sub>*, and *r*<sub>33</sub> by *r*<sub>13</sub>.

To effect a polarization transformation, other elements of the electro-optic tensor are utilized. The EO tensor of LiNbO<sub>3</sub> has the following form:

$$[r] = \begin{bmatrix} 0 & -r_{22} & r_{13} \\ 0 & r_{22} & r_{13} \\ 0 & 0 & r_{33} \\ 0 & r_{51} & 0 \\ r_{51} & 0 & 0 \\ -r_{22} & 0 & 0 \end{bmatrix}. \quad (2)$$

The three columns of the tensor represent the effects of an applied field along the *x*, *y*, and *z*-axes, respectively. The first three rows represent changes in *n<sub>xx</sub>*, *n<sub>yy</sub>*, and *n<sub>zz</sub>*, respectively (e.g., *r*<sub>33</sub> is related to the change in *n<sub>zz</sub>* due to *E<sub>z</sub>*). The remaining three rows are related to the rotations of the principal axes of the crystal about the *x*, *y*, and *z*-axes, respectively. Thus *r*<sub>61</sub> (which is numerically equal to  $-r_{22}$ )

\*Address all correspondence to James E. Toney, E-mail: jtoney@srco.com



**Fig. 1** (a) A center-drive configuration in x-cut, z-propagating device for TE–TM conversion, (b) differential-drive configuration for frequency shifting, and (c) z-cut, x-propagating device for PPLN TE–TM converter.

indicates a rotation of the crystal's  $xy$ -axes due to a field applied along the  $x$ -axis, while  $r_{42}(=r_{51})$  gives a rotation of the  $yz$ -axes due to  $E_y$ . These last two tensor elements are utilized in polarization-transforming devices.

## 2.2 Polarization Transformations in Z-Propagating LiNbO<sub>3</sub>

The simplest geometry for a guide-wave electro-optic polarization transformer in LiNbO<sub>3</sub> is a straight waveguide with the propagation direction along the crystallographic  $z$ -axis. Since the  $z$ -axis is the symmetry axis of the crystal, there is no material birefringence for light polarized in the  $x-y$  plane. There may, however, be waveguide birefringence, which can typically be cancelled by a DC bias voltage.

If a uniform electric field is applied along the crystal's  $y$ -axis, the primary effect is to induce a birefringence between the TE and TM modes of

$$\Delta n_{\text{TE-TM}} = \gamma n_o^3 r_{22} E_y. \quad (3)$$

The electro-optic phase retardation between the two modes over an interaction length,  $L$ , is given as

$$\Gamma = \gamma n_o^3 r_{22} E_y \frac{2\pi}{\lambda} L. \quad (4)$$

This phase retardation can induce a transformation between linear, elliptical, and circular polarization states.

A field applied along the  $y$ -axis cannot effect a conversion between TE and TM modes, since they are polarized along the principal axes of the crystal and are therefore eigenmodes of the Jones matrix. For mode conversion, the electric field is applied along the crystallographic  $x$ -axis, inducing a rotation of the  $xy$ -axes by  $\sim 45^\circ$  deg. (The precise angle is slightly different due to the effect of  $r_{51}$ .) The birefringence and phase retardation between waves polarized along the rotated  $x$ - and  $y$ -axes are given in Eqs. (3) and (4), with  $E_y$  replaced by  $E_x$ .

In the rotated principal-axis system, the TE and TM modes are represented in the Jones vector formalism as

$$V_{\text{TE}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad (5a)$$

$$V_{\text{TM}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (5b)$$

Under a phase retardation  $\Gamma$ , the Jones vectors become

$$V_{\text{TE}} \rightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} e^{-i\Gamma/2} \\ -e^{i\Gamma/2} \end{pmatrix}, \quad (6a)$$

$$V_{\text{TM}} \rightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} e^{-i\Gamma/2} \\ e^{i\Gamma/2} \end{pmatrix}. \quad (6b)$$

For  $\Gamma = \pi$ , the TE mode converts completely to TM, and vice versa:

$$V_{\text{TE}} \xrightarrow{\Gamma=\pi} \frac{1}{\sqrt{2}} \begin{pmatrix} -i \\ -i \end{pmatrix} = -iV_{\text{TM}}, \quad (7a)$$

$$V_{\text{TM}} \xrightarrow{\Gamma=\pi} \frac{1}{\sqrt{2}} \begin{pmatrix} -i \\ i \end{pmatrix} = -iV_{\text{TE}}. \quad (7b)$$

Both types of polarization transformation can be realized in a single optical waveguide placed beneath the center electrode of a RF CPW. In a  $x$ -cut crystal,  $E_x$  is applied by driving the center electrode (or equivalently, by grounding the center electrode and driving the two outer electrodes symmetrically), while  $E_y$  is applied by driving the two outer electrodes asymmetrically.

## 2.3 TE-to-TM Conversion in Periodically Poled Lithium Niobate

An alternative device structure for implementing TE–TM conversion is based on periodically poled lithium niobate (PPLN). This structure employs the  $r_{42}(=r_{51})$  element of the EO tensor to effect a small rotation of the principal axes about the  $x$ -axis, by application of an electric field along the  $y$ -axis. Devices based on this effect have been demonstrated using interdigitated electrodes, as in Refs. 1–3, or by periodic poling (PPLN).<sup>8,9,10</sup>

A coupled-mode theory (CMT) analysis of TE–TM conversion by a periodic reversal of the electro-optic effect was published in the early 1970s.<sup>11</sup> An intuitive way of understanding the operation of this device is to view each domain segment as a half-waveplate, which transforms light that is linearly polarized at an angle  $+\theta$  relative to its principal axes into  $-\theta$  (i.e., it rotates the polarization by an angle  $2\theta$ ). The next PPLN segment, having its optic axis reversed, rotates the polarization by  $6\theta$  in the opposite direction, for a net rotation of  $4\theta$ . The process continues in the periodic structure, with the cumulative polarization rotation after  $N$  periods given by  $4N\theta$ . The number of periods for complete TE–TM conversion is, therefore, given as

$$N_{TE-TM} = \frac{\pi}{8\theta}, \quad (8)$$

where  $\theta$  is the rotation of the crystal's principal axes due to the electro-optic effect under an applied field amplitude  $E_y$ , which has been shown to be

$$\theta \approx \frac{r_{51}E_y}{\frac{1}{n_e^2} - \frac{1}{n_o^2}}. \quad (9)$$

A requirement for this device to operate as described is that the optical path difference between  $e$  and  $o$  waves as they pass through one domain segment is equal to one-half of the free-space wavelength  $\lambda$ . That is, the period  $\Pi$  of the PPLN structure is given as

$$\Pi \approx \frac{\lambda}{n_o - n_e}. \quad (10)$$

Thus, a device with a given poling period operates in a narrow wavelength band, which can be adjusted by temperature tuning. Combining Eqs. (8), (9), and (10), the required field for TE–TM conversion for a device of length  $L$  is

$$E_y \approx \frac{\pi\lambda\bar{n}}{4n_e^2n_o^2r_{51}L}, \quad (11)$$

where  $\bar{n}$  is the mean of  $n_e$  and  $n_o$ . (In the CMT result,  $\bar{n}$  is the geometric mean rather than the arithmetic mean, but the numerical difference is negligible.)

From Eq. (10), it can be observed that the conversion wavelength for a given period is proportional to the birefringence. Since the refractive indices depend on temperature, the wavelength can be tuned by thermoelectric heating or cooling. Accounting for both thermo-optic and thermal expansion effects, the sensitivity with temperature is

$$\frac{\partial\lambda}{\partial T} = \Pi \left( \frac{\partial n_o}{\partial T} - \frac{\partial n_e}{\partial T} \right) + (n_o - n_e) \frac{\partial \Pi}{\partial T}. \quad (12)$$

The thermo-optic effect is an order of magnitude larger than the thermal expansion effect; back-of-the-envelope calculations give a sensitivity of  $-0.11^\circ\text{C}/\text{nm}$  for the thermo-optic term, and  $+0.015^\circ\text{C}/\text{nm}$  for the thermal expansion term.

This calculated thermo-optic effect does not account for the pyroelectric effect, which causes a further increase in birefringence due to the induced field along the  $z$ -axis and the associated electro-optic effect. Consequently the measured wavelength shift with temperature (see Sec. 4.3) is  $\sim 4$  times larger than expected from the simple calculation.

### 3 Multiphysics Modeling

COMSOL Multiphysics 4.2 finite element modeling (FEM) software was used to construct an integrated model of both optical and RF waveguides. COMSOL enables integrated simulation of the waveguide diffusion processes, DC or RF electric fields, and optical modes.<sup>12,13</sup> The dopant profile of the ZnO-diffused waveguide was calculated with a time-dependent diffusion model using the transport of dilute species interface. The RF and optical modes, and the coupling

between them, were calculated using the electromagnetic waves interface.

### 3.1 Multiphysics Model Construction

#### 3.1.1 Diffusion model

The dopant source was modeled as a thin region of unity initial concentration above the surface of the LiNbO<sub>3</sub> wafer over a width equal to the lithographically defined waveguide width.

The effective diffusion coefficient was calculated from a published temperature-dependent equation of the form

$$D_e(T) = D_0 e^{-E/kT}, \quad (13)$$

where  $E$  is the activation energy,  $T$  is the absolute temperature, and  $k$  is the Boltzmann constant. The diffusion time and temperature are set to match the fabrication process. Model parameters, based on prior published work,<sup>14</sup> were  $E = 3.0$  eV and  $D_0 = 0.00608$  m<sup>2</sup>/s.

#### 3.1.2 RF model

The electromagnetic mode solver was used to compute the mode(s) of a CPW structure at frequencies from 1.0 to 10.0 GHz. The gold electrodes were modeled as regions of finite conductivity with a dielectric constant of 1. Integration couplings within COMSOL were used to compute the current,  $I$ , and average power,  $P$ , as in Ref. 13. The characteristic impedance was then calculated from

$$Z_0 = \frac{2P}{|I|^2}. \quad (14)$$

The RF loss coefficient was calculated from the imaginary part of the effective index as

$$\alpha = -\frac{2\pi f \text{Im}(n_{\text{eff}})}{c}. \quad (15)$$

#### 3.1.3 Optical mode calculation

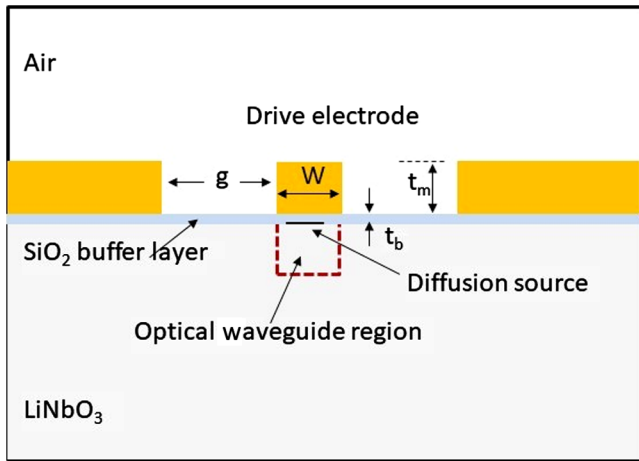
The LiNbO<sub>3</sub> substrate indices ( $n_e$  and  $n_o$ ) at the operating wavelength  $\lambda$  were calculated from the published Sellmeier relation.<sup>15</sup> For diffused waveguides, the refractive index change was taken as proportional to the dopant distribution from step 1. The **electro-optic refractive index change** is calculated based on the RF mode computed in step 2 from

$$\Delta n = \pm \frac{1}{2} n^3 r_{22} E_i, \quad (16)$$

where  $E_i$  is the relevant RF electric field component, and  $n$  is either the ordinary ( $n_o$ ) or extraordinary ( $n_e$ ). The  $+$  and  $-$  signs apply to orthogonal (TE versus TM) modes.

#### 3.1.4 Calculation of electro-optic overlap

Since the computed RF mode is not normalized, the voltage  $V$  must be extracted from the computed field for normalization. This is done within COMSOL via a line integral of the tangential electric field component along a line between the ground and center electrodes of the CPW. For a specified



**Fig. 2** Model geometry for CPW.

**Table 1** Design parameters and material properties for CPW model.

| Parameter                                | Symbol       | Value             | Units         |
|------------------------------------------|--------------|-------------------|---------------|
| Electrode thickness                      | $t_m$        | 7                 | $\mu\text{m}$ |
| Center electrode width                   | $W$          | 8                 | $\mu\text{m}$ |
| Electrode gap                            | $G$          | 15                | $\mu\text{m}$ |
| Buffer thickness                         | $t_b$        | 1                 | $\mu\text{m}$ |
| LiNbO <sub>3</sub> relative permittivity | $\epsilon_r$ | 78, 78, 32        |               |
| SiO <sub>2</sub> relative permittivity   | $\epsilon_r$ | 3.8               |               |
| Au conductivity                          | $\sigma$     | $4.5 \times 10^7$ | S/m           |

device length  $L$ , the half-wave voltage is then calculated as

$$V_\pi = \frac{V\lambda}{2\Delta n_m L}, \quad (17)$$

where  $\Delta n_m$  is the electro-optic-induced birefringence between TE and TM modes. This value of  $V_\pi$  does not include the effect of RF loss.

### 3.2 RF Properties

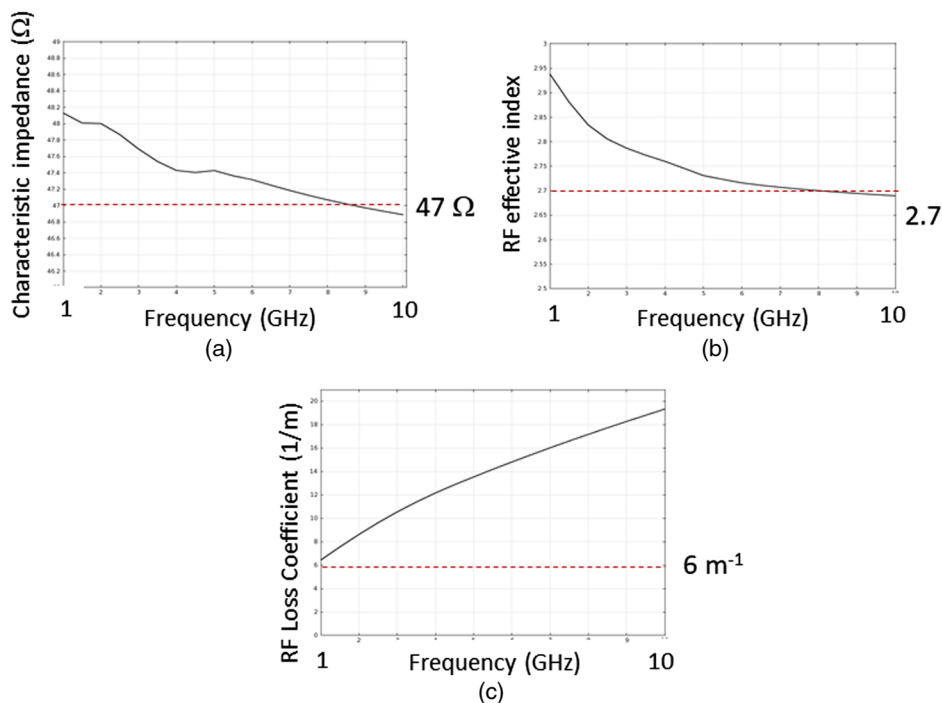
COMSOL was used to compute the characteristic impedance, effective index, and loss coefficient as a function of frequency for the CPW configuration. The coplanar stripline configuration of Fig. 1(c) was not simulated, as it was intended for low-speed operation.

The model geometry is illustrated in Fig. 2. A region of high mesh resolution was included to enable accurate calculation of the optical modes.

The electrode dimensions were chosen to achieve impedance and velocity matching by a manual iteration process rather than a formal optimization algorithm. Design parameters and assumed material properties are given in Table 1. The characteristic impedance, effective index, and RF loss coefficient versus frequency from 1 to 10 GHz are shown in Fig. 3. The impedance is closely matched to  $50\Omega$ , and the RF effective index is less than 3, enabling velocity matching to a frequency of several gigahertz.

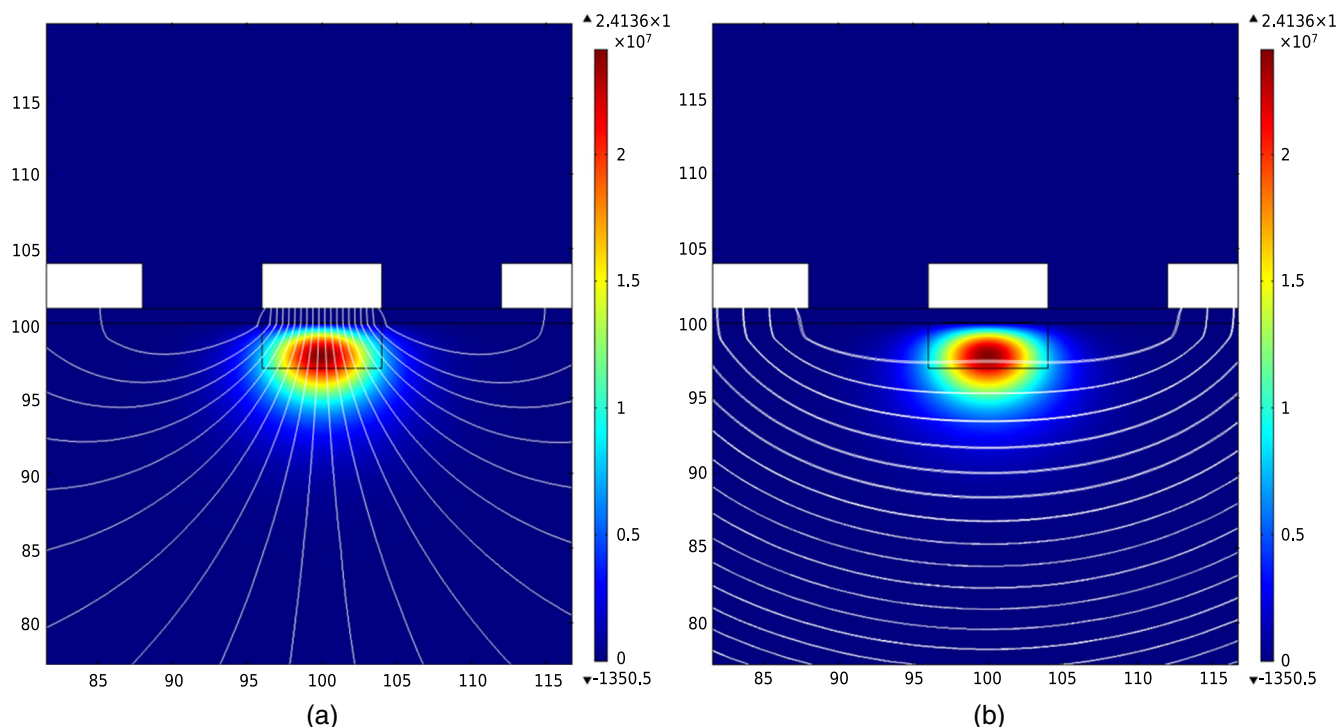
### 3.3 Optical and Electro-Optic Properties

Optical TE and TM modes were computed at a wavelength of 780 nm, first with electro-optic coefficient set to 0 to determine the waveguide birefringence, then with  $r_{22}$  set to the



**Fig. 3** Computed RF properties of coplanar waveguide from 1 to 10 GHz. (a) Characteristic impedance, (b) effective index, and (c) loss coefficient.





**Fig. 4** Computed optical (TE) mode overlaid with RF electric field lines. (a) Antisymmetric mode corresponding to center-drive configuration. (b) Symmetric mode corresponding to differential drive. Dimensions are in microns.

published value (3.4 pm/V for RF frequencies) to determine the electro-optically induced birefringence. The TE mode, overlaid with electric field lines for both the center-drive and differential-drive configurations, is shown in Fig. 4. The waveguide width of 7  $\mu\text{m}$  was chosen based on prior experience, in which it was found to give the best combination of low propagation loss, good coupling efficiency to an external fiber, and single-mode operation.

From the computed effective indices of the TE and TM modes, the waveguide birefringence is computed to be  $5.3 \times 10^{-5}$ , and the electro-optic-induced birefringence as  $1.0 \times 10^{-6}/\text{V}$ . From Eq. (4), the overlap factor is calculated as  $\gamma = 0.38$ . The TE–TM conversion voltage at 780 nm is then, from Eq. (17), 40 V/cm, or 13 V for a 3-cm-long device. Based on the RF and optical effective indices, the velocity-matching bandwidth is given as

$$f_c L = \frac{c}{2n_{RF} \left(1 - \frac{n_o}{n_{RF}}\right)} = 23 \text{ GHz-cm}. \quad (18)$$

## 4 Experimental Results

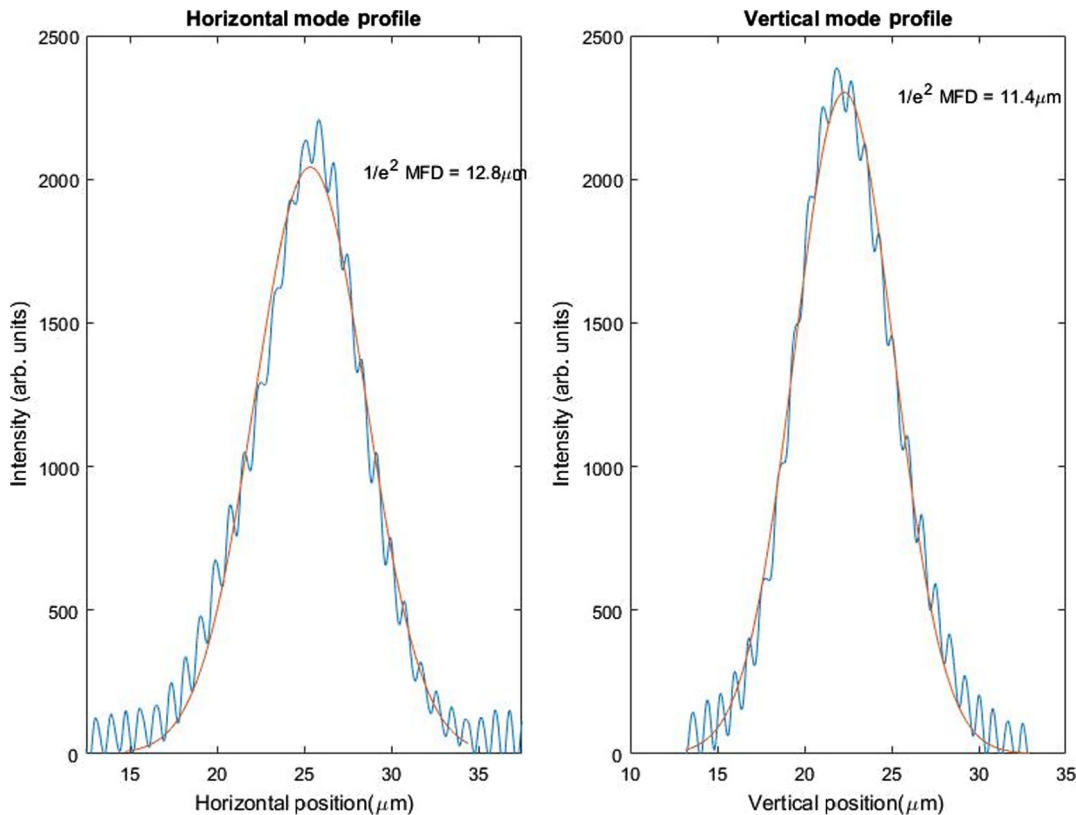
### 4.1 Device Fabrication

Optical waveguides were fabricated by in-diffusion of zinc oxide into the  $x$ - or  $z$ -face (depending on device type) of the lithium niobate wafer. Standard photolithography was used to define a 6- to 7- $\mu\text{m}$ -wide waveguide opening. A zinc oxide layer was deposited, and the wafer was annealed in an oxygen atmosphere. As with any diffusion process, there was a range of functionally equivalent combinations of time and

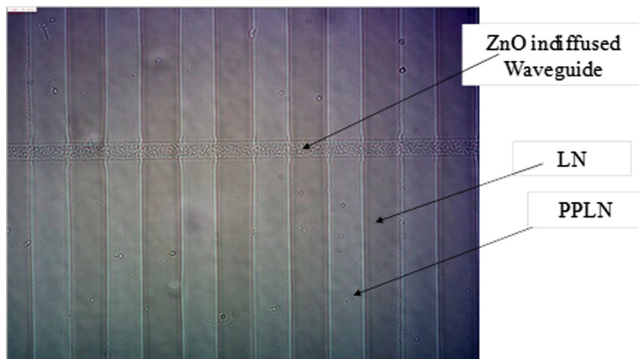
temperature, based on the published temperature-dependent diffusion coefficient.<sup>14</sup>

The optical mode field at 1064 nm is measured using a DataRay profiler. The measured horizontal and vertical profiles for the TM mode are shown in Fig. 5 with a Gaussian curve fit. The oscillations in the measured profile are due to interference effects. The ZnO waveguide mode profile is less elliptical than other diffused waveguides in LiNbO<sub>3</sub>, such as annealed proton exchange (APE) or Ti-diffused waveguides. The vertical profile is more symmetrical than APE or Ti waveguides, with less of a tail on the substrate side. These characteristics are helpful in achieving high input/output coupling efficiency.

Periodic poling for the PPLN device was carried out using a liquid-electrode poling method, as described in Ref. 12. The periodic pattern was defined by standard photolithography on the  $+z$  crystal face, and the photoresist was hard-baked to serve as an insulator. The poling period was calculated from Eq. (10) to be 14.0  $\mu\text{m}$  for a wavelength of 1064 nm at room temperature. The length of each opening in the resist was chosen to achieve a duty cycle near 50% in the periodic domain pattern, allowing for spreading of domains during the poling process. The wafer was sealed in a cell with layers of aqueous salt solution covering both surfaces. A high-voltage amplifier applied a voltage between the  $+z$  and  $-z$  faces greater than the coercive field of  $\sim 20,000 \text{ V/mm}$ . The poling current was monitored and the process terminated when the integrated current reached the expected value, given the total area being inverted and the known spontaneous polarization of LiNbO<sub>3</sub>, of 72  $\mu\text{C}/\text{cm}^2$ . A photo of a periodically poled section and waveguide is shown in Fig. 6.



**Fig. 5** Measured mode profile of TM mode for a 6- $\mu\text{m}$  ZnO:LiNbO<sub>3</sub> waveguide at 1064 nm with Gaussian curve fits. Interference fringes due to reflections are apparent in the measurements.

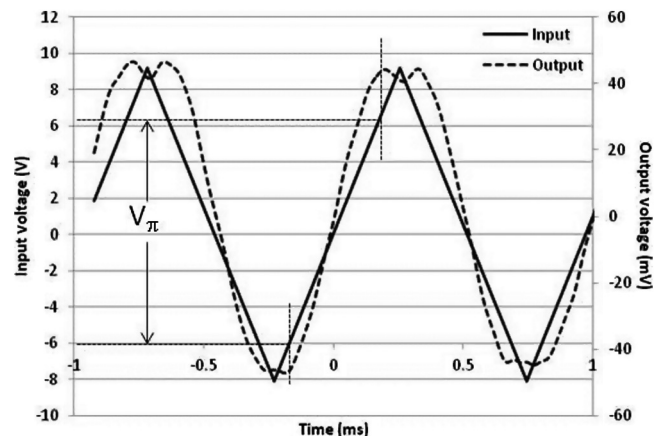


**Fig. 6** Periodically poled section and ZnO-diffused waveguide.

Gold electrodes with a Ti adhesion layer were defined lithographically and applied by electron-beam evaporation. For the z-propagating device, the electrode structure was a CPW with the optical waveguide under the center conductor. For the PPLN device, it was a coplanar stripline with the waveguide between the two conductors.

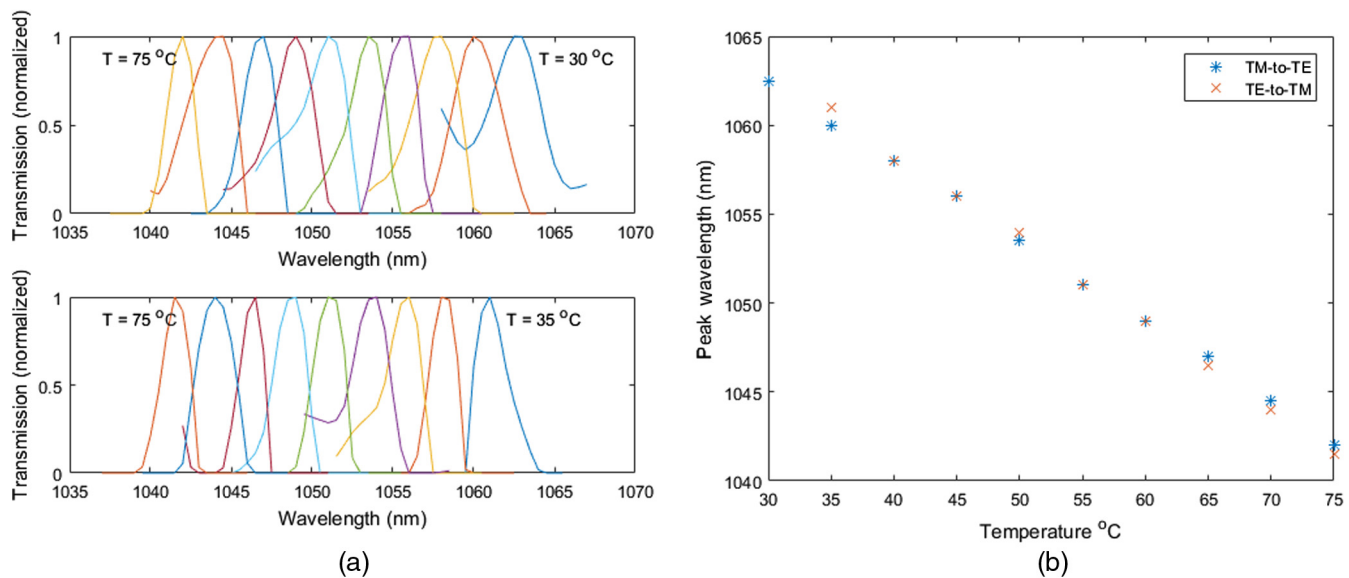
#### 4.2 Z-Propagating TE-TM Converter

An approximate value of the half-wave voltage was determined by driving the electrodes with a 1-kHz triangle waveform and increasing the amplitude until the output peak-to-peak amplitude reached its maximum value. The input and output waveforms at the onset of saturation are shown in Fig. 7. Since the needle probe arrangement is susceptible



**Fig. 7** Determination of half-wave voltage from low-frequency response.

to pickup of external noise, the output waveform was digitally low-pass filtered with a cut-off frequency of 40 kHz. The approximate value of  $V\pi$  is 12 V. From Eq. (16), the product  $\gamma r_{22} E_x$  can be estimated as  $1.1 \times 10^{-6}$  (dimensionless), given the active device length of 3 cm. Using as a nominal value for  $E_x$ , the voltage divided by the electrode gap of 15  $\mu\text{m}$  gives  $\gamma r_{22} \approx 1.7 \text{ pm/V}$ . The published low-frequency value of  $r_{22}$  for lithium niobate is 6.8 pm/V, which leads to  $\gamma \approx 0.25$ . This value is approximately two-thirds of the prediction of the FEM model, implying that the fabrication process and electrode design can be improved to achieve a significant reduction in operating voltage.



**Fig. 8** (a) Spectral response of PPLN polarization rotator at various temperatures and (b) peak wavelength versus temperature. Some of the structures of the peaks are due to imperfect background subtraction related to variation in the state of polarization of the SLD source.

### 4.3 Periodically Poled Lithium Niobate TE–TM Converter

A periodically poled LiNbO<sub>3</sub> device with a poling period of 14  $\mu$ m, for operation at a wavelength of 1064 nm, was fabricated. The device was integrated with a thermoelectric heater for temperature tuning of the resonant wavelength. A 1060-nm superluminescent diode (SLD) source was coupled into the input, and an in-line polarizer was placed at the output, oriented orthogonally to the dominant input polarization. The spectrum of the transmitted light was observed with a fiber optic spectrometer.

A voltage was applied to the electrodes to rotate the input polarization by 90 deg, producing a transmission peak at the resonant wavelength. The measurement was repeated as the device temperature was tuned in 5°C steps from 30°C to 75°C. The transmitted spectra after background subtraction are shown in Fig. 8(a) for both TM-to-TE and TE-to-TM conversions. Figure 8(b) shows the peak wavelength versus temperature for conversion in each direction. The temperature sensitivity from a least-squares fit, combining both sequences into one data set, is  $-0.46$  nm/°C.

## 5 Conclusions

Polarization-transforming devices have been demonstrated in ZnO:LiNbO<sub>3</sub> waveguides. TE–TM conversion has been demonstrated in two configurations: a  $z$ -propagating geometry and an  $x$ -propagating periodically poled (PPLN) structure. The PPLN device has a narrow-band wavelength response that was swept over the range of 1040 to 1064 nm by temperature tuning from 30°C to 75°C.

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**James E. Toney** is a senior research scientist at SRICO, Inc. and associated faculty at Ohio State University. He received his BS degree in electrical engineering from Rensselaer Polytechnic Institute, New York, in 1984, his MS degree in physics from Indiana University of Pennsylvania in 1994, and his PhD in applied physics from Carnegie Mellon University, Pennsylvania, in 1998. His research

interests include electric field sensors and other electro-optic devices. He is a senior member of SPIE and IEEE.

**Jason Retz** is a research engineer at SRICO. He earned his BS and MS degrees in engineering physics from Ohio State University.

**Andrea Pollick** is the director of engineering at SRICO. She earned her BS degree in biomedical engineering from Marquette University, Wisconsin.

**Vincent E. Stenger** is the director of research and development at SRICO. He earned his BS, MS, and PhD degrees in electrical engineering from the University of Cincinnati, Ohio.

**Sri Sriram** is the president and founder of SRICO. He earned his PhD in electrical engineering from the University of Cincinnati in 1980.